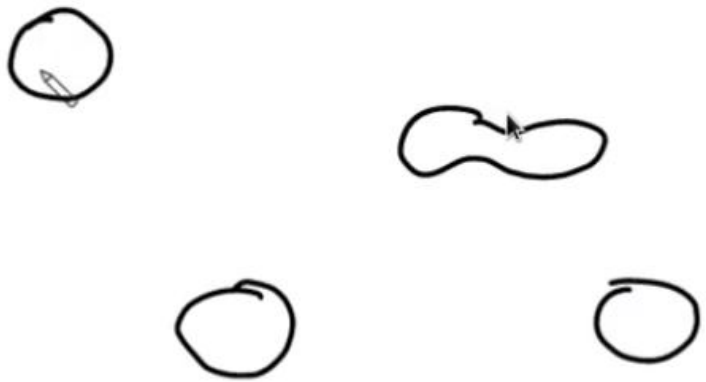
LIVE

Limit of Resolution / Resolving power





Aviraj Roy

AV Aviraj Roy
exactly

KA Kartik Kumar
2 light

HA Harshal Sethiya
Phir maut dikhegi

AV Aviraj Roy
exactly !!

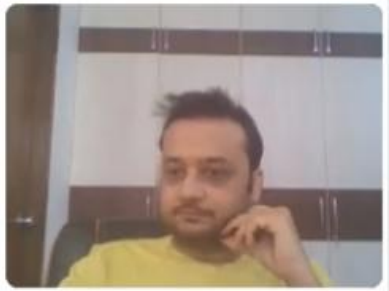
PR Pratham omkar Pat...
Haha

SR Srijahnavi Ch
😄

Type a message

Chat Doubts More





Pratham Omkar Pat...
Haha

SR Srijahnavi Ch
😂

HA Harsh Dixit dixit
Yy

Abhiharsh Chauhan
😂

SO Soumya Thakur
can we say the same for distant building or city light at night?...when viewed from terrace

Type a message

Chat Doubts More

LIVE

Limit of Resolution / Resolving power

Resolving power of any optical instrument is the capability to form distinct images of closeby objects.



building or city

light at night?...when viewed from terrace

KA Kartik Kumar: webb

HA Harsh Kanu: Jwst

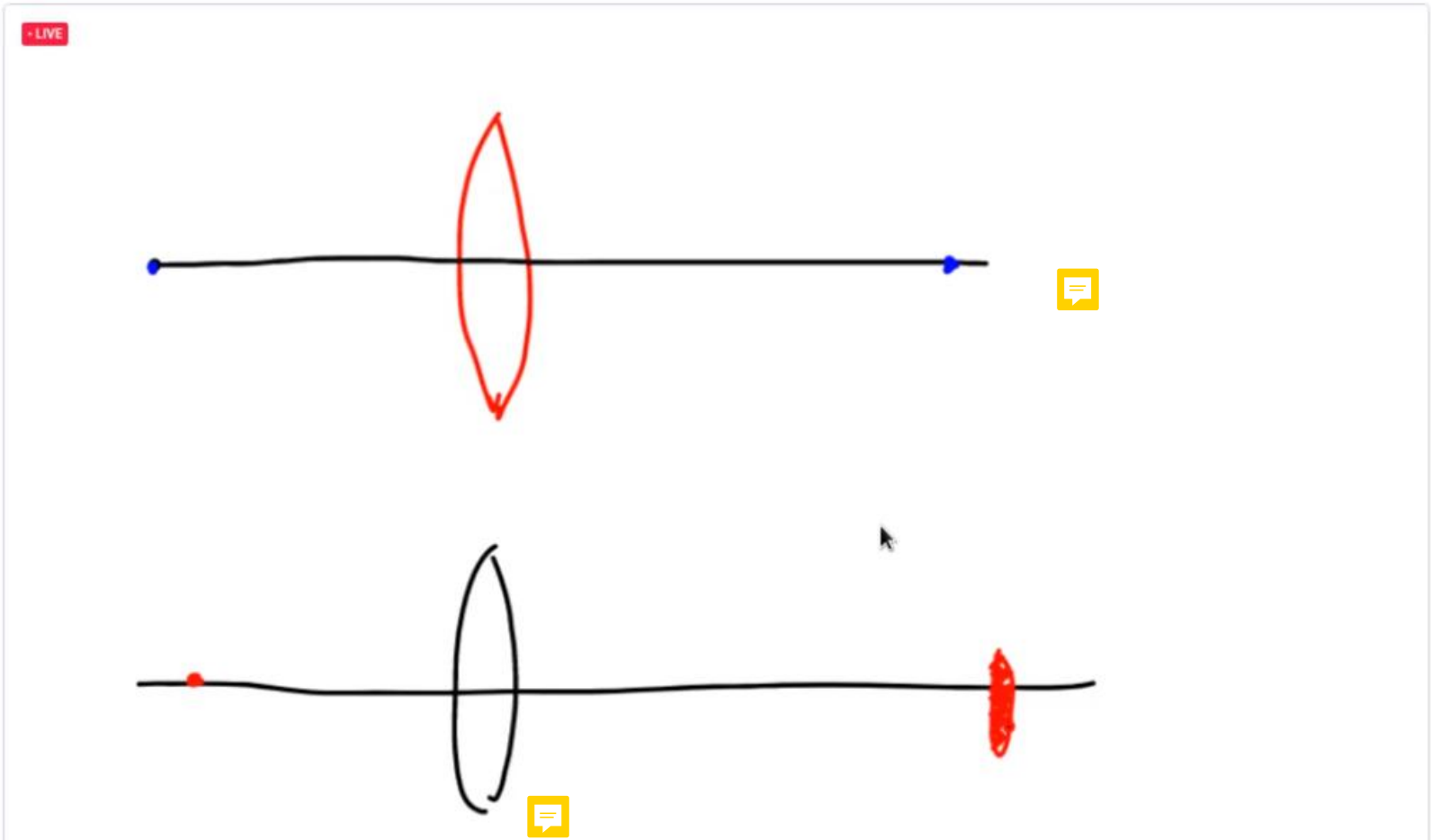
VA Vasu Khanna: JWST

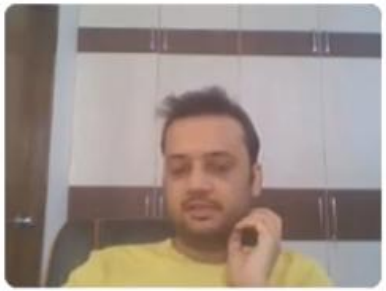
AV Aviraj Roy: webb

KA Kartik Kumar: james webb'

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Chat Doubts More





JWST

AV Aviraj Roy
webb

KA Kartik Kumar
james webb'

HA Harshal Sethiya
Replay dekhega toh 😂

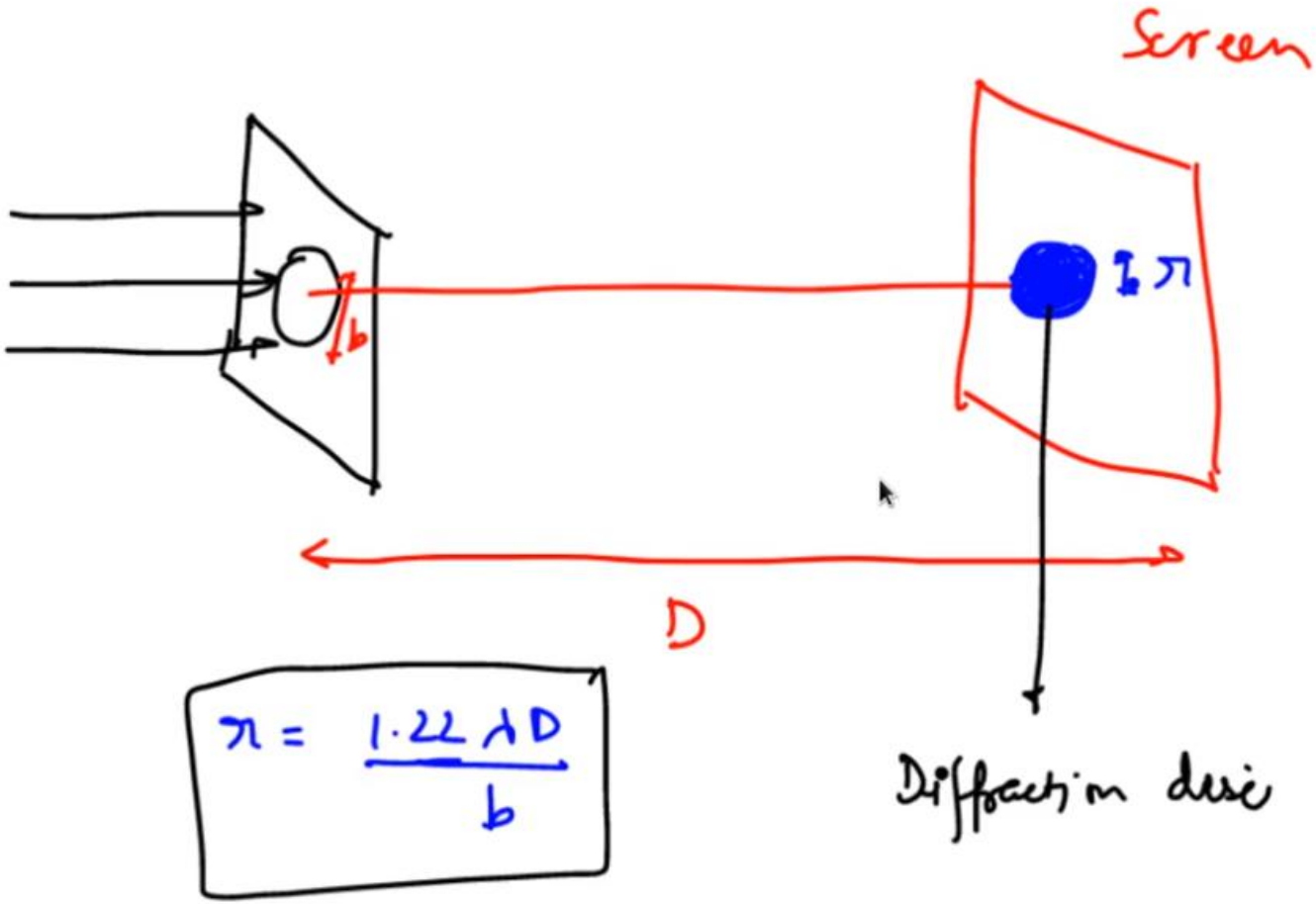
A A Hari kumar
haha

HA Harsh Kanu
XD

AV Aviraj Roy
hehe

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Chat Doubts More



$$x = \frac{1.22 \lambda D}{b}$$



HA Harsh Kanu
.

AB Abhiharsh Chauhan
y

PR Pratham omkar Pat...
y

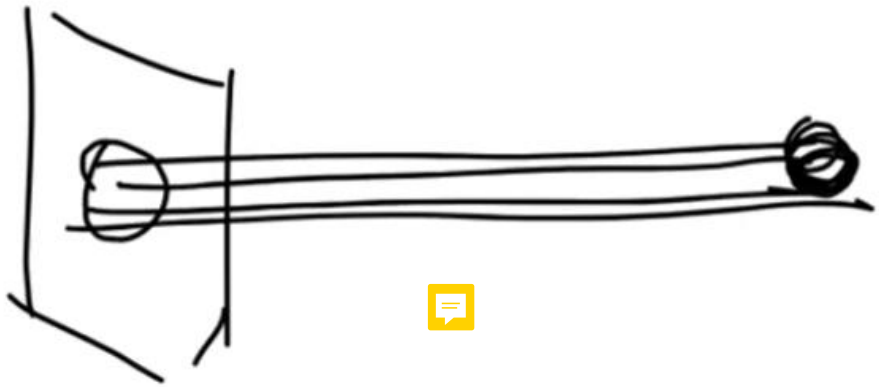
KA Kartik Kumar
yes

JA Jashan Grover Insan
i performed this at home after reading this for the first time 😀 😀

Type a message

Chat Doubts More

LIVE





HA Harsh Kanu
.

AB Abhiharsh Chauhan
y

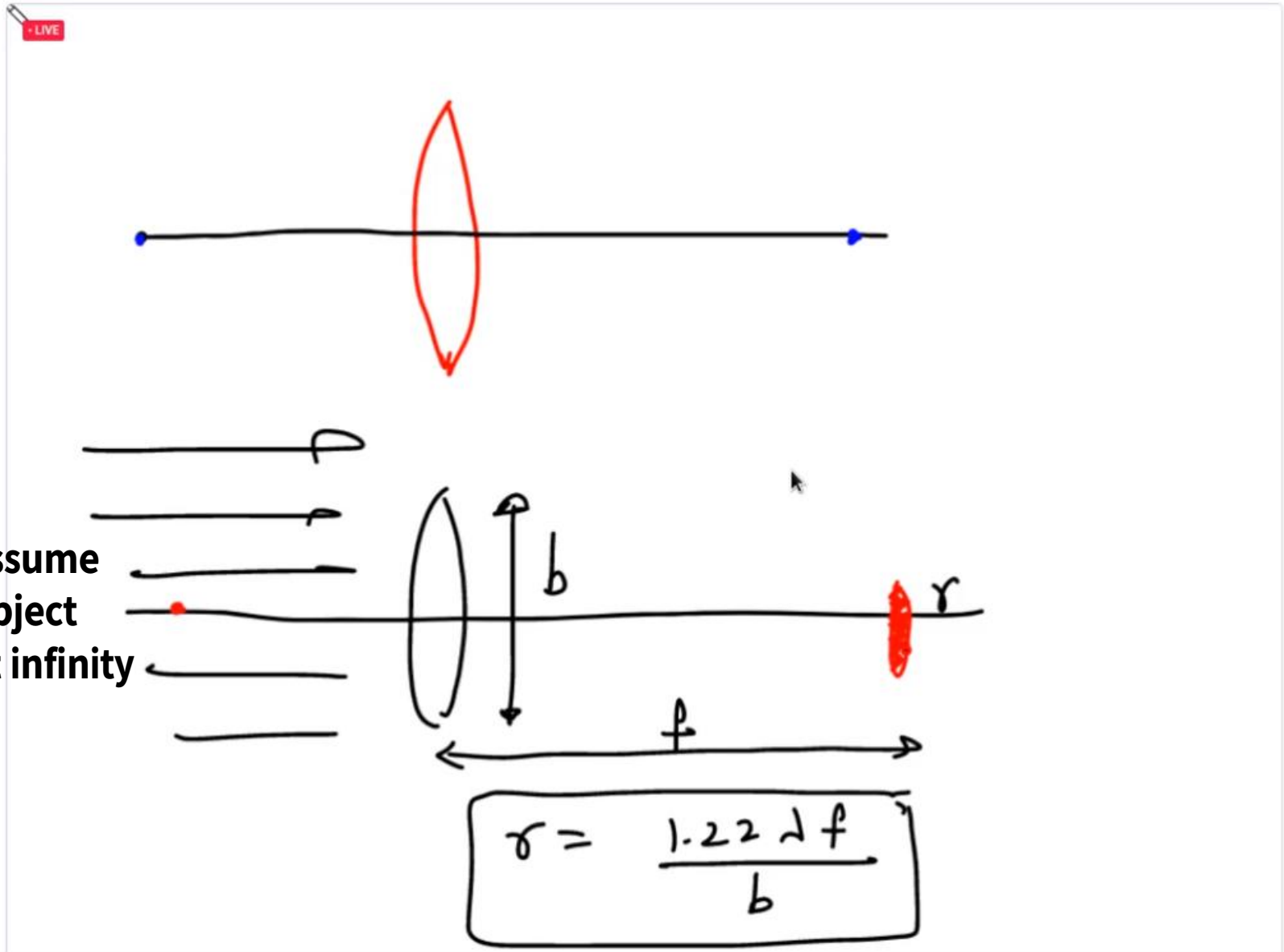
PR Pratham omkar Pat...
y

KA Kartik Kumar
yes

JA Jashan Grover Insan
i performed this at home after reading this for the first time 😁😁

Type a message

Chat Doubts More





HA Harsh Kanu
.

AB Abhiharsh Chauhan
y

PR Pratham omkar Pat...
y

KA Kartik Kumar
yes

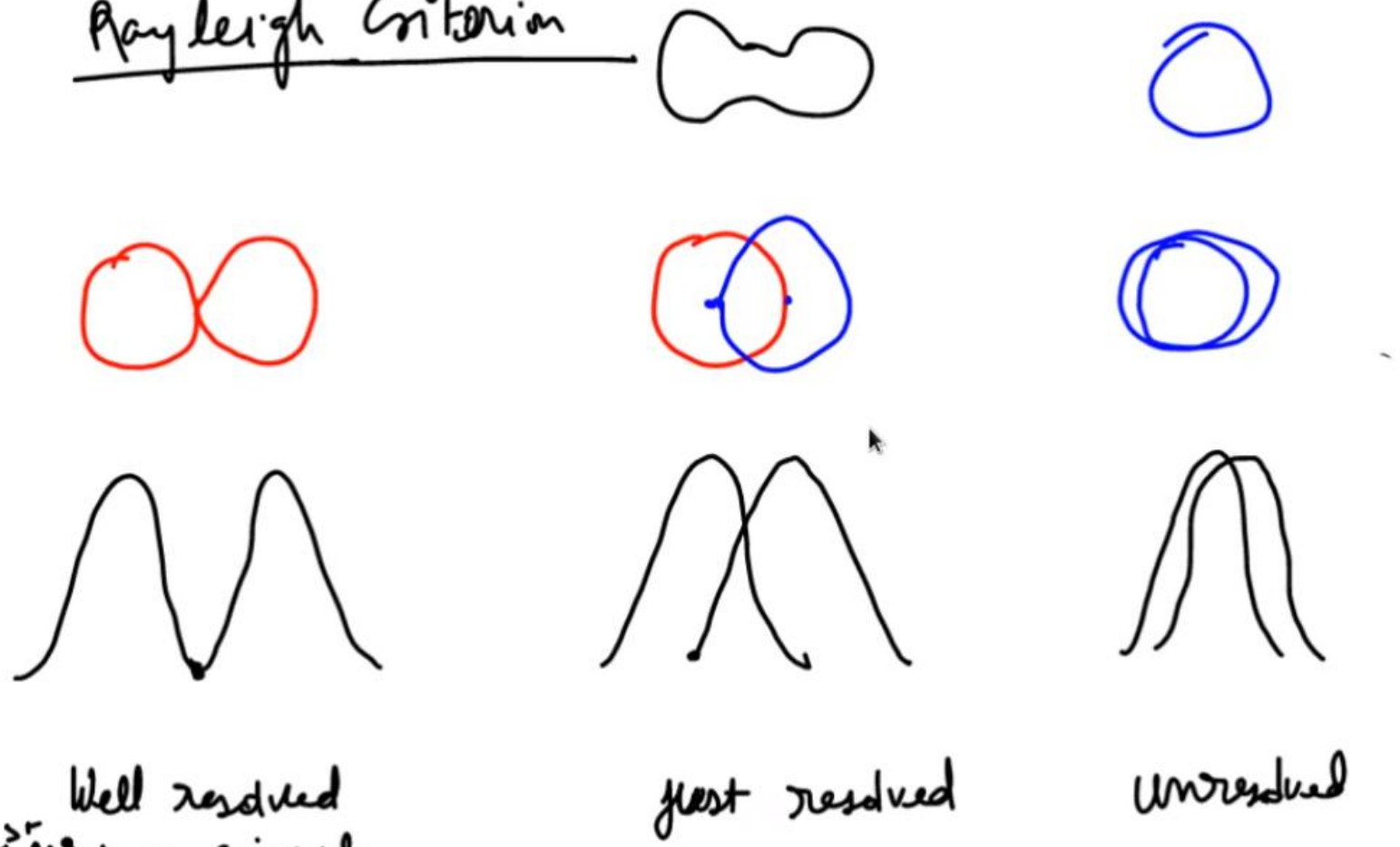
JA Jashan Grover Insan
i performed this at home after reading this for the first time 😀 😀

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LIVE

Rayleigh Criterion



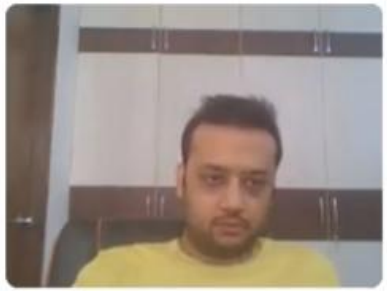


A Hari Kumar
all
HA Harsh Kanu
all
VA Vasu Khanna
pcm
AV Aviraj Roy
curcor holi khelne gaya hai
KA Kartik Kumar
undissolvedd
Srijahnavi Ch
SR Srijahnavi Ch
yes
Type a message
Chat
Doubts
More

LIVE

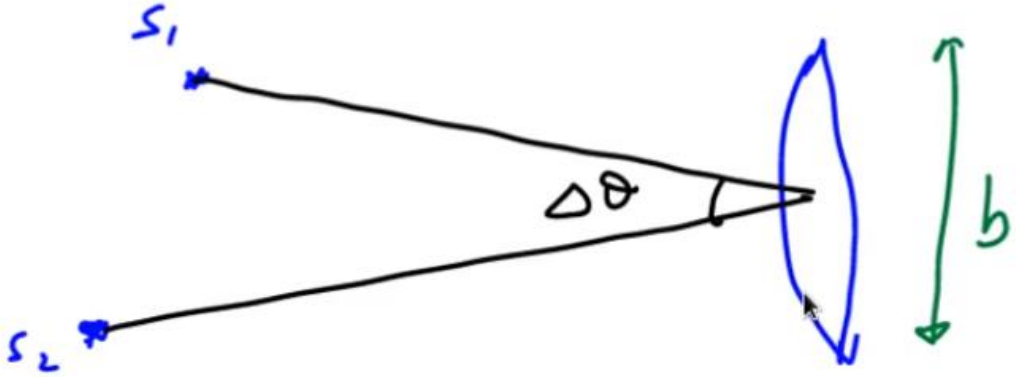
Rayleigh Criteria

For the objects to be just resolved, the maxima of first should coincide with the minima of second & vice versa



LIVE

Resolving power of a telescope



Limit of Resolution of Telescope is defined as minimum value of $\Delta\theta$ such that S_1 and S_2 can be just resolved

$$\Delta\theta = \frac{1.22\lambda}{b}$$



KA Kartik Kumar hubble

JA Jashan Grover Insan hubble

SR Srijahnavi Ch y

AV Aviraj Roy hubble

HA Harsh Kanu H

SA Sanyam kumar Jain 1

PR Prathik y

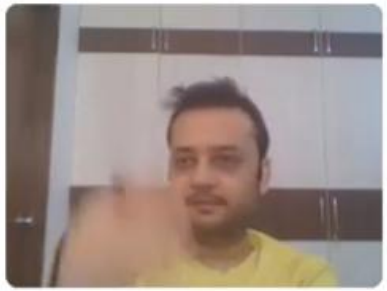
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Chat Doubts More

λ = wavelength of the light

b = diameter of the aperture

$$RP = \frac{1}{RL} = \frac{1}{\Delta\theta} = \frac{b}{1.22\lambda}$$



KA Kartik Kumar
china

KA Kartik Kumar
yes

KA Kartik Kumar
south m h kahi

VA Vasu Khanna
LBT

RI Rishav Raj
india ke paas total 14 telescope hain

RI Rishav Raj
operational

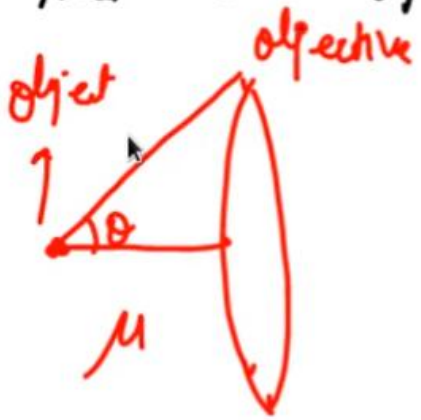
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Chat Doubts More

LIVE

Microscope Resolving Power / Resolution Limit

Limit of Resolution of a microscope is defined as the minimum distance such that the object can be just resolved



$$d_{min} = \frac{1.22\lambda}{2\mu \sin\theta}$$

μ = RI of the medium b/w object & objective

λ = wavelength of light

$\mu \sin\theta$ = Numerical Aperture of the microscope



hai

HA Harshal Sethiya
Koi ek baat pe sehmat kyo nahi hota

SR Srijahnavi Ch
ok

KA Kartik Kumar
itna approx ni krenge verma ji

AV Aviraj Roy
truth is ncert sabka baap hai

AB Abhiharsh Chauhan
y

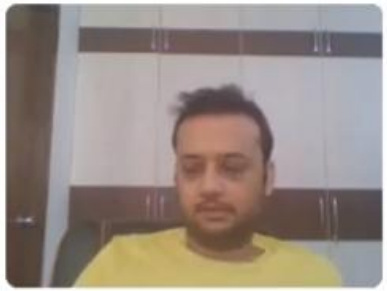
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Chat Doubts More

LIVE

$$d_{\min} = \frac{1.22\,\lambda}{2n\sin\beta}$$

$$d_{min} = \frac{\lambda}{2\mu \sin\beta}$$



hai

HA Harshal Sethiya
Koi ek baat pe sehmat kyo nahi hota

SR Srijahnavi Ch
ok

KA Kartik Kumar
itna approx ni krenge verma ji

AV Aviraj Roy
truth is ncert sabka baap hai

AB Abhiharsh Chauhan
y

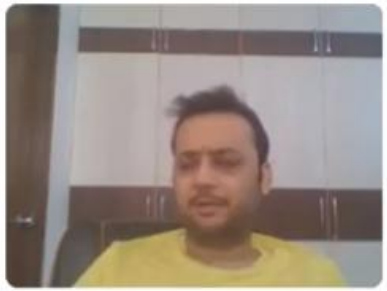
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LIVE

$$RP = \frac{1}{d_{min}} = \frac{2\mu \sin\theta}{1.22\lambda}$$

⇒ Generally oil is kept between object & objective to increase the resolving power



Sanyam kumar Jain
y

A Hari kumar
y

Soumya Thakur
so if we take denser medm RP is more

Soumya Thakur
okk

Shardul Date
resolving power = 1/ (resolving limit) this is by definition?

Type a message

Chat Doubts More

LIVE

Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm coming from a distant object, the limit of resolution of the telescope is close to: [2019]

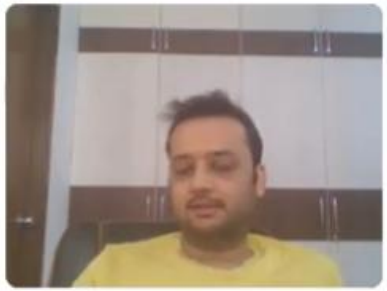
(a) 1.5×10^{-7} rad (b) 2.0×10^{-7} rad

(c) 3.0×10^{-7} rad (d) 4.5×10^{-7} rad

$b = 2.5$

$\lambda = 600\ nm$

$\frac{1.22\lambda}{b} = \Delta\theta$



AV Aviraj Roy
c

HA Harshal Sethiya
Yahan hum logo ko thodi jhel rhe hote

AB Abhiharsh Chauhan
c

HA Harsh Kanu
c

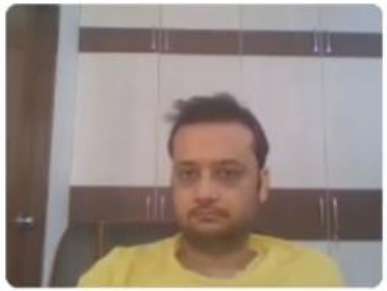
HA Harshal Sethiya
Jio jio bol ke zindagi haram kr dete

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Chat Doubts More

LIVE

Q: Two stars are both at a distance of 10^{15} m from Earth. The separation b/w them is d. They are viewed by a telescope of aperture radius 0.5 m. If the wavelength of the light is $\lambda = \frac{10^{-7}}{1.22}$ m then find the minimum possible distance b/w the stars so that they can be viwed as two different stars.



zindagi naram kr

dete

RI Rishav Raj
10^8

PR Prathik
10^8 m

A A Hari kumar
2x 10^-2 m

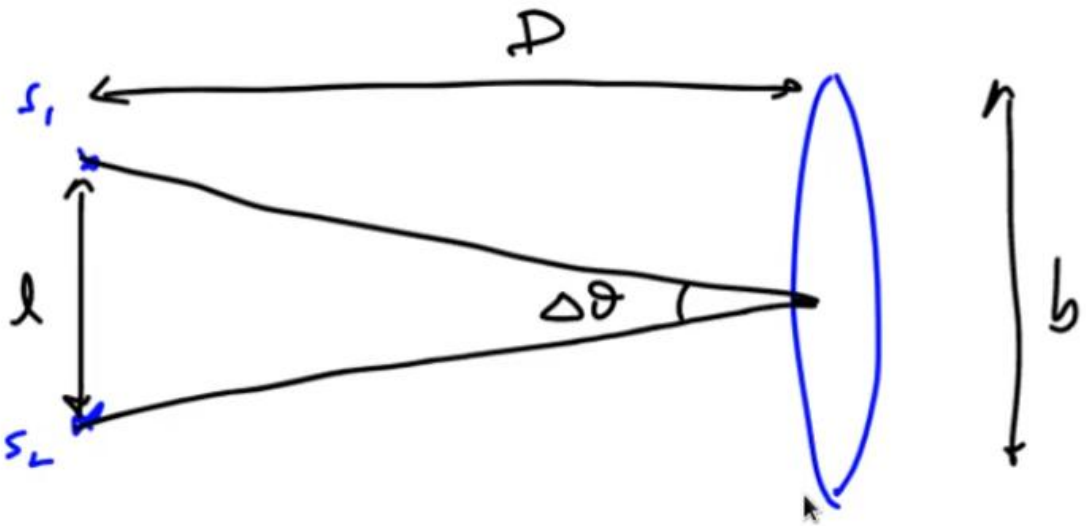
AB Abhiharsh Chauhan
2*10^8

HA Harsh Kanu
2 10^8

SA Sanyam kumar Jain
10^8

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Chat Doubts More



$$\Delta\theta_{min} = \frac{1.22\lambda}{b} = \frac{l}{D}$$

$$= \frac{1.22}{1.22} \times \frac{1}{1} \times \frac{10^{-7}}{10^{15}}$$

$$l = 10^8$$



AV Aviraj Roy
10'8

AV Aviraj Roy
*^

PR Pratham omkar Pat...
10^8

A A Hari kumar
2 x 10^8

KA Kartik Kumar
same as hari

KA Kartik Kumar
yes

HA Harsh Kanu
ok

Type a message

Chat Doubts More